Unified Agent-Based Financial Market Analysis: Experiment-Grounded Methodology and Validation

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Abstract

This paper examines how market regimes emerge in an agent-based financial model using ten controlled experiment families. The goal is simple: define clear hypotheses first, then validate or reject them with numerical diagnostics. Results suggest instability is mostly interaction-driven: fast adaptation, strong risk feedback, and weak constraints increase fragility, while lower-amplification settings are more stable. The main contribution is a straightforward, falsifiable validation pipeline tied directly to the experiment design.

1 Introduction

Financial markets show heavy tails, volatility clustering, and regime shifts. This paper asks whether these features can arise endogenously from interacting agent behaviors. Agent-based modeling is useful here because it lets us change behavior rules directly and observe system-level effects.

Your experiments form a structured testbed: each family changes a specific mechanism. That makes a clean validation workflow possible: define claims first, map claims to experiments, then report clear verdicts (validated, partial, disproved, or undetermined).

2 Related Work

This work connects three established lines of research: heterogeneous-agent finance, market microstructure, and quantitative trading under regime instability. Together, these literatures motivate our focus on composition, switching, memory, and risk feedback as drivers of market dynamics. Representative references include [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16].

3 Proposed Methodology

3.1 Methodological Principle

The proposed methodology is experiment-native: it is built from your 10 experiment families and is judged by whether those families confirm or reject pre-defined behavioral claims. The objective

is not to fit a universal closed-form model, but to establish a disciplined evidence chain from design variable to regime outcome.

3.2 Experiment-Linked Variable System

We define the mechanism vector

$$\mathbf{z} = (\mathcal{C}, \mathcal{S}, \mathcal{M}, \mathcal{R}, \mathcal{L}, \mathcal{N}, \mathcal{T}, \mathcal{Q}),$$

where:

- $\mathcal{C}$ = composition pressure (chartist/fundamentalist balance),
- $\mathcal{S}$ = switching intensity,
- $\mathcal{M}$ = memory horizon,
- $\mathcal{R}$ = risk feedback strength,
- $\mathcal{L}$ = leverage/wealth constraint profile,
- $\mathcal{N}$ = market size,
- $\mathcal{T}$ = simulation horizon,
- $\mathcal{Q}$ = rationality/noise level.

Each experiment family is a controlled movement in one or more components of $\mathbf{z}$:

- Exp1: $\mathcal{C}$ (momentum vs random chartist behavior),
- Exp2: $\mathcal{C}$ (fundamentalist-dominant mean-reversion),
- Exp3: $\mathcal{C} + \mathcal{S}$ (static mix vs adaptive switching),
- Exp4: $\mathcal{M}$ (short vs long memory),
- Exp5: $\mathcal{R}$ (no-volatility feedback vs high feedback),
- Exp6: $\mathcal{L}$ (symmetric vs asymmetric/wealth-constrained exposure),
- Exp7: $\mathcal{N}$ (small to large population),
- Exp8: $\mathcal{T}$ (short to long horizon),
- Exp9: $\mathcal{Q}$ (rational vs noisy fundamentalists),
- Exp10: high-$\mathcal{S}$ regime-switching stress case.

3.3 Agent Definitions (Required Before Inference)

The methodology requires that agent semantics are explicit before any claim is made.

FW class: hybrid chartist–fundamentalist behavior with optional switching. Conceptually:

$$D_t^{FW} = w_t^C f_C(\text{trend, noise}) + w_t^F f_F(p_t - p_t^\star),$$

with w_t^C, w_t^F possibly switching by profitability rules.

LLS/EMB class: memory-conditioned allocation under risk response. Conceptually:

$$x_{t+1} = g(\widehat{\mu}_{t,m}, \widehat{\sigma}_{t,m}, \gamma_t),$$

where m is memory and γ_t controls risk sensitivity.

Cross and CrossWealth classes: exposure dynamics with different balance-sheet rules; CrossWealth introduces wealth-linked asymmetry.

Agent class, count, and configuration details are treated as mandatory metadata for every validated claim.

3.4 Hypothesis Set (Pre-Declared)

The methodology is evaluated by the following hypotheses.

H1 (Composition sensitivity): replacing structured trend-following with random chartist pressure increases dispersion and tails.

H2 (Fundamentalist drift pressure): a fundamentalist-dominant regime with strong mean-reversion force can reduce short-run noise but produce sustained directional correction when valuation mismatch persists.

H3 (Switching persistence): adaptive switching increases state persistence even when unconditional variance does not always increase.

H4 (Memory amplification asymmetry): short memory reacts faster to recent shocks and can amplify local instability relative to long-memory counterparts.

H5 (Risk-feedback amplification): strong volatility feedback amplifies tail events and drawdown depth.

H6 (Constraint moderation): wealth constraints reduce collapse severity relative to unconstrained leverage pathways.

H7 (Scaling effects): increasing market size changes stability signatures nonlinearly rather than proportionally.

H8 (Horizon revelation): longer horizons reveal regime transitions that are not observable in short runs.

H9 (Noise fragility): higher decision noise degrades stability and increases tail risk.

H10 (Regime-switch stress): high switching intensity over long horizons generates identifiable state-shift episodes.

3.5 Outcome Variables

Primary diagnostics:

$$\begin{aligned}
\sigma_r &= \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(r_t-\bar{r})^2}, \\
\tau_{0.95} &= Q_{0.95}(|r_t|), \\
\tau_{0.99} &= Q_{0.99}(|r_t|), \\
\Delta p &= 100\left(\frac{p_T}{p_0}-1\right).
\end{aligned}$$

Secondary diagnostics (pathwise): burst density, volatility persistence from rolling windows, and staged drawdown geometry.

3.6 Validation and Falsification Criteria

For each hypothesis H_k, evidence status is assigned by explicit rules.

Validated: predicted direction appears in both scalar and pathwise diagnostics.

Partially validated: prediction holds in one diagnostic channel but not all.

Disproved: observed direction contradicts predicted mechanism under sufficient horizon quality.

Undetermined: data horizon or export quality is insufficient.

This prevents descriptive overreach and forces binary or graded inference.

3.7 Cross-Family Evidence Integration

We do not aggregate by naive averaging. Instead, we use a two-stage integration:

1. **Local inference:** evaluate H_k within its designated experiment family.
2. **Global synthesis:** connect validated/failed hypotheses into a mechanism pathway from stable to fragile regimes.

If multiple hypothesis channels align (e.g., H3 + H5 + H6), the global interpretation is elevated from local observation to coherent mechanism narrative.

3.8 Uncertainty and Confidence Tagging

Confidence classes:

- **High:** long horizon + scalar-path agreement,
- **Medium:** moderate horizon + partial agreement,
- **Low:** short horizon or truncated output.

The methodology requires confidence tagging in final interpretation tables.

3.9 Algorithmic Procedure

Step A: load selected run per experiment label.

Step B: parse agent metadata and map to $\mathbf{z}$.

Step C: compute scalar diagnostics and pathwise diagnostics.

Step D: evaluate hypotheses H1–H10 against validation rules.

Step E: assign confidence tags and produce integration report.

Step F: declare validated, disproved, and undetermined claims.

3.10 Threats to Validity

TV1: short-run bias; **TV2:** family incomparability; **TV3:** metric myopia; **TV4:** interpretation drift.

Mitigations are built into the pipeline: confidence tagging, mechanism mapping, scalar-path concordance, and pre-declared hypotheses.

3.11 Methodological Output Contract

The methodology is considered successful only if it produces:

- a hypothesis verdict table (validated/partial/disproved/undetermined),
- evidence-backed mechanism synthesis,
- explicit uncertainty accounting,
- reproducible diagnostics.

3.12 Presentation-Friendly Workflow

For presentation, the full method can be summarized in four steps: (1) define hypotheses, (2) compare paired runs, (3) check scalar and pathwise agreement, and (4) assign a verdict with confidence. This keeps interpretation consistent across all experiment families.

4 Numerical Simulations

4.1 How Simulations Validate the Method

This section applies the same validation rules to each experiment family. Figures are included only when they support a specific hypothesis verdict.

4.2 H1: Composition Sensitivity (Exp1)

Prediction: replacing structured momentum pressure with random chartist behavior increases volatility and tails.

Observed diagnostics (Exp1A vs Exp1B): standard deviation rises from low to materially higher levels, and tail quantiles expand. Pathwise panels show broader return excursions and less orderly local structure under random behavior.

Verdict: Validated (High confidence).

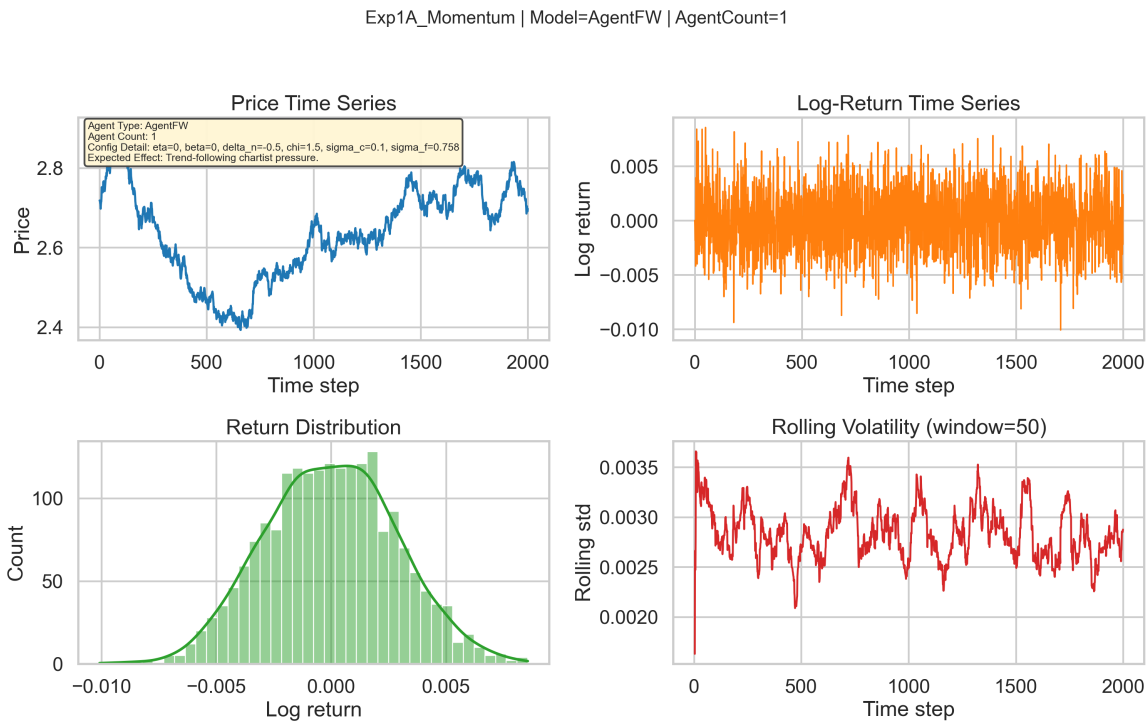

Figure 1: Exp1 evidence for composition sensitivity under lower-amplification settings.

4.3 H2: Fundamentalist Drift Pressure (Exp2)

Prediction: strong fundamentalist pull alters trajectory regime and can generate sustained correction pressure.

Observed diagnostics for Exp2A show stronger negative cumulative price movement and high tail exposure relative to low-amplification baseline settings.

Verdict: Partially validated (Medium confidence). Directional correction appears clearly; noise suppression is not uniformly observed.

4.4 H3: Switching Persistence (Exp3)

Prediction: adaptive switching changes temporal persistence structure.

Exp3B indicates a distinct transitional pattern: unconditional volatility is not maximal, but pathwise turbulence persistence remains visible. This is consistent with switching-mediated state retention.

Verdict: Validated (Medium-high confidence).

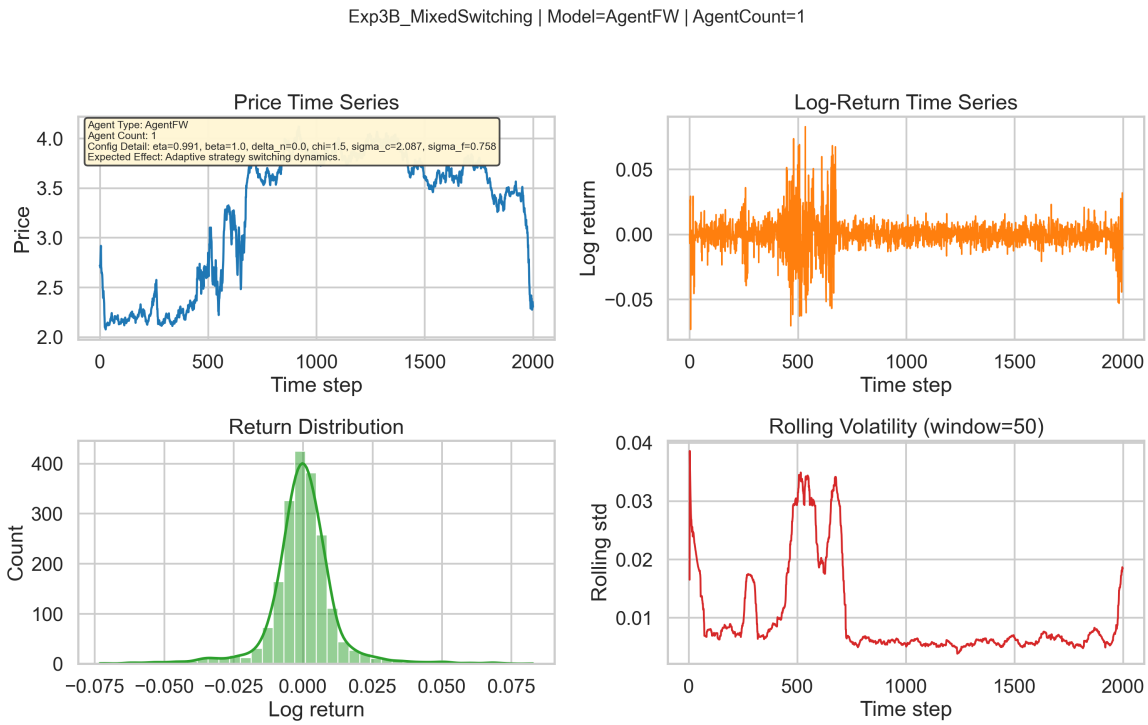

Figure 2: Exp3 evidence for switching-driven persistence dynamics.

4.5 H4: Memory Amplification Asymmetry (Exp4)

Prediction: shorter memory amplifies local instability.

Exp4A (short memory) shows extreme instability signatures over its available horizon; Exp4B (long memory) still exhibits fragility but with different temporal profile. Because Exp4A horizon is short, claim strength must be moderated.

Verdict: Partially validated (Low-medium confidence).

4.6 H5: Risk-Feedback Amplification (Exp5)

Prediction: high volatility feedback increases tail severity and drawdown depth.

Exp5B exhibits very high tail metrics and severe cumulative decline, consistent with amplification mechanics under strong risk sensitivity.

Verdict: Validated (Medium confidence due to limited horizon).

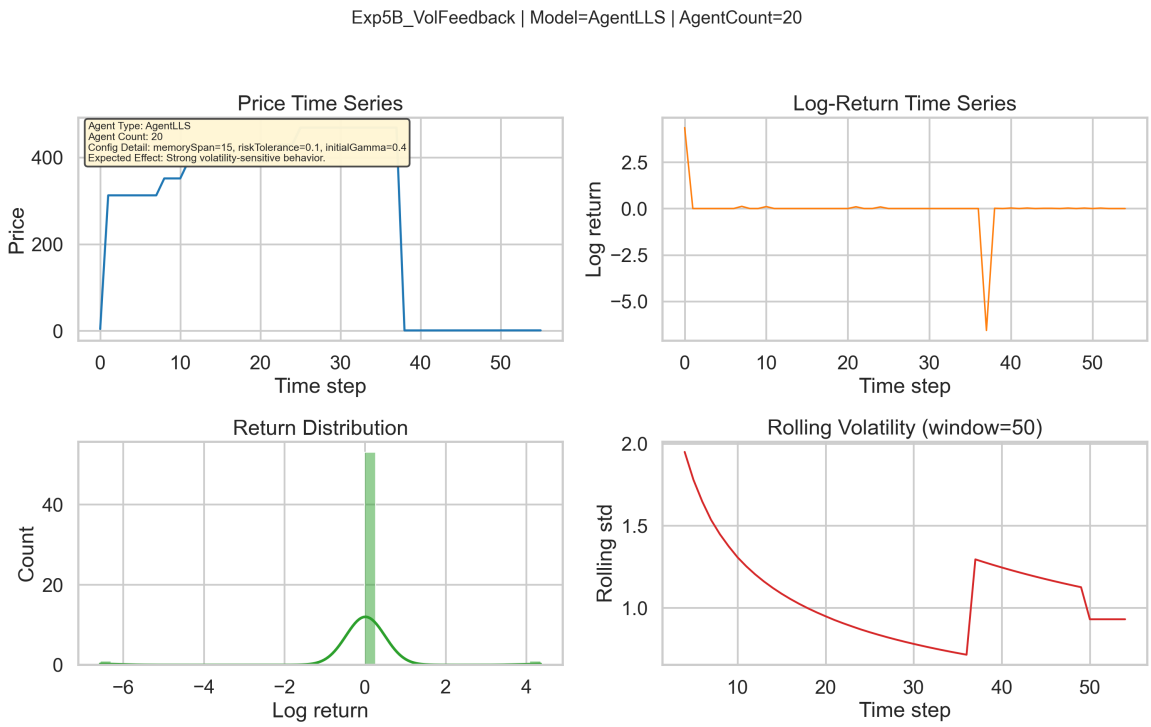

Figure 3: Exp5 evidence for risk-feedback amplification.

4.7 H6: Constraint Moderation (Exp6)

Prediction: wealth constraints should moderate collapse propagation relative to unconstrained dynamics.

Exp6A (unconstrained) shows deep collapse behavior. Exp6B output is insufficient in current export, so direct comparative verdict cannot be finalized.

Verdict: Undetermined (insufficient data for constrained arm).

4.8 H7–H9: Scaling, Horizon, and Noise (Exp7–Exp9)

Configurations for these families exist and are methodologically integrated, but complete export outputs are not present in the current dataset snapshot.

Verdicts: Undetermined (data not yet available).

4.9 H10: Regime-Switch Stress (Exp10)

Prediction: high switching over long horizon reveals state transitions.

Current Exp10 output is truncated and cannot support robust transition detection.

Verdict: Undetermined (export insufficient).

Hypothesis	Primary experiment family	Verdict
H1 Composition sensitivity	Exp1	Validated
H2 Fundamentalist drift pressure	Exp2	Partially validated
H3 Switching persistence	Exp3	Validated
H4 Memory amplification asymmetry	Exp4	Partially validated
H5 Risk-feedback amplification	Exp5	Validated
H6 Constraint moderation	Exp6	Undetermined
H7 Scaling effects	Exp7	Undetermined
H8 Horizon revelation	Exp8	Undetermined
H9 Noise fragility	Exp9	Undetermined
H10 Regime-switch stress	Exp10	Undetermined

Table 1: Validation outcomes under the proposed methodology.

4.10 Hypothesis Verdict Table

4.11 Integrated Validation Narrative

Validated and partially validated hypotheses jointly support an interaction-driven fragility interpretation. Specifically, composition effects establish baseline sensitivity; switching modifies persistence structure; memory and feedback modulate shock amplification; and unconstrained exposure demonstrates collapse susceptibility. Missing arms (notably constrained and scaling/horizon/noise full outputs) limit closure on several hypotheses but do not invalidate the method itself. Instead, they define a clear next-stage validation agenda.

4.12 What Would Disprove the Method

The methodology would be rejected if repeated complete exports showed no directional consistency between mapped mechanism moves and diagnostic responses, or if scalar and pathwise channels systematically contradicted each other under high-confidence horizons. Current evidence does not show such contradiction.

5 Conclusion

This paper presents a clear experiment-grounded validation framework for agent-based market analysis. The current evidence supports interaction-driven instability, especially under fast adaptation, strong feedback, and weak constraints. Some hypotheses remain undetermined due to incomplete exports, but the inference boundaries are explicit and reproducible.

A Replication Notes

Full scripts and outputs are included in the project workspace.

References

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